ELECTRICAL AND ELECTRONICS ENGINEERING

Complete 50-Page Course Textbook

Beginner → Intermediate

Circuit Theory • Electronics • Digital Systems • Machines • Measurements • Safety • Control

A structured course covering fundamental electrical quantities, circuit analysis, electronic components, digital logic, measurements, electrical machines, power electronics, control systems, microcontrollers, PCB design, safety, and troubleshooting.

Course Overview

Purpose: This course introduces learners to the principles and practical concepts used in electrical and electronics engineering.

Learning outcomes: Learners should be able to analyze basic circuits, identify components, explain semiconductor operation, understand digital logic, use measurement instruments safely, describe electrical machines, understand control systems, and apply systematic troubleshooting.

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Lesson 1: Introduction to Electrical and Electronics Engineering

Lesson Content

Electrical engineering deals broadly with electrical energy, circuits, machines, power, control, and related systems. Electronics focuses strongly on devices and circuits that control, process, amplify, switch, or transmit electrical signals.

Exercise

Differentiate electrical engineering from electronics engineering.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 2: Electrical Quantities and Units

Lesson Content

Voltage is electrical potential difference, current is the movement of electric charge, resistance opposes current, and power represents the rate of electrical energy transfer. Common units include volts, amperes, ohms, watts, and joules.

Exercise

Define voltage, current, resistance, and power.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 3: Charge, Current and Voltage

Lesson Content

Electric charge is measured in coulombs. Current is charge flow rate, while voltage represents potential difference that can drive current through a circuit.

Exercise

Explain the relationship among charge, current, and voltage.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 4: Ohm's Law

Lesson Content

Ohm's law relates voltage, current, and resistance using V = IR for an ohmic component under appropriate conditions. It is fundamental to circuit analysis.

Exercise

Use Ohm's law to find an unknown electrical quantity.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 5: Electrical Power and Energy

Lesson Content

Electrical power can be calculated from voltage and current, with useful forms including $P = VI$, $P = I^2R$, and $P = V^2/R$ when applicable. Energy is power multiplied by time.

Exercise

Explain the difference between electrical power and electrical energy.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 6: DC and AC

Lesson Content

Direct current has a consistent direction of flow, while alternating current periodically changes direction. DC is common in electronic systems and batteries; AC is widely used in power distribution.

Exercise

Give three differences between AC and DC.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 7: Series Circuits

Lesson Content

In a series circuit, components share a single current path. The same current flows through each series element, while voltage divides among components according to circuit conditions.

Exercise

State three characteristics of series circuits.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 8: Parallel Circuits

Lesson Content

Parallel branches share the same two principal nodes, so branch voltage is common while current divides among branches.

Exercise

State three characteristics of parallel circuits.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 9: Kirchhoff's Laws

Lesson Content

Kirchhoff's Current Law expresses charge conservation at a node, while Kirchhoff's Voltage Law expresses energy conservation around a closed loop.

Exercise

Explain KCL and KVL.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 10: Circuit Analysis

Lesson Content

Circuit analysis uses systematic methods to determine currents, voltages, and power. Techniques include Ohm's law, Kirchhoff's laws, node analysis, and mesh analysis.

Exercise

Outline a basic circuit-analysis procedure.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 11: Resistors

Lesson Content

Resistors limit current and create voltage drops. Important properties include resistance value, tolerance, power rating, and physical construction.

Exercise

Explain the purpose of a resistor and its power rating.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 12: Resistor Color Codes

Lesson Content

Through-hole resistors may use colored bands to indicate resistance and tolerance. Reading the code requires understanding significant digits, multiplier, and tolerance conventions.

Exercise

Practice reading a standard resistor color code.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 13: Potentiometers and Variable Resistors

Lesson Content

Variable resistors allow resistance or voltage division to be adjusted. Potentiometers are commonly used as adjustable voltage dividers and control inputs.

Exercise

Give three applications of variable resistors.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 14: Capacitors

Lesson Content

Capacitors store energy in an electric field and oppose rapid changes in voltage. Capacitance is measured in farads and depends on construction and dielectric properties.

Exercise

List four applications of capacitors.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 15: Capacitor Charging and Discharging

Lesson Content

In RC circuits, capacitor voltage changes over time rather than instantaneously. The time constant is related to resistance and capacitance.

Exercise

Explain the meaning of an RC time constant.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 16: Inductors

Lesson Content

Inductors store energy in magnetic fields and oppose changes in current. They are used in filters, converters, transformers, and many power circuits.

Exercise

List four applications of inductors.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 17: Transformers

Lesson Content

Transformers transfer AC energy between circuits through electromagnetic induction. They can step voltage up or down and provide isolation in suitable designs.

Exercise

Explain the operating principle of a transformer.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 18: Diodes

Lesson Content

A diode is a semiconductor device that preferentially conducts in one direction under normal operation. Diodes are used in rectification, protection, switching, and signal processing.

Exercise

Name five applications of diodes.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 19: LEDs

Lesson Content

Light-emitting diodes produce light when appropriately forward biased. They are efficient indicators and light sources but require suitable current control.

Exercise

Why does an LED normally require current limiting?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 20: Rectifiers

Lesson Content

Rectifier circuits convert AC into pulsating DC. Half-wave and full-wave arrangements differ in how much of the input waveform is used.

Exercise

Differentiate half-wave and full-wave rectification.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 21: Zener Diodes

Lesson Content

Zener diodes are designed to operate in a controlled reverse-breakdown region for applications such as voltage reference and regulation.

Exercise

Give two uses of Zener diodes.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 22: Transistors

Lesson Content

Transistors are semiconductor devices used for switching, amplification, and signal control. Common families include bipolar junction transistors and field-effect transistors.

Exercise

State three applications of transistors.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 23: BJT Fundamentals

Lesson Content

A BJT has three terminals commonly called emitter, base, and collector. Its operating behavior depends on biasing and configuration.

Exercise

Name the three terminals of a BJT.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 24: MOSFET Fundamentals

Lesson Content

A MOSFET uses an electric field at its gate to control conduction between its main current terminals. MOSFETs are widely used in switching and power electronics.

Exercise

Give four applications of MOSFETs.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 25: Operational Amplifiers

Lesson Content

An operational amplifier is a high-gain differential amplifier commonly used with feedback to perform mathematical and signal-processing functions.

Exercise

List five applications of operational amplifiers.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 26: Feedback and Amplifiers

Lesson Content

Feedback returns part of a circuit output to an input path. Negative feedback can improve stability and control gain, bandwidth, and distortion characteristics.

Exercise

Explain negative feedback in an amplifier.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 27: Digital Electronics

Lesson Content

Digital systems represent information using discrete logic states, commonly described as binary 0 and 1. Logic circuits form the basis of computers and digital controllers.

Exercise

Differentiate analog and digital signals.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 28: Number Systems

Lesson Content

Binary, decimal, hexadecimal, and sometimes octal systems are used in electronics and computing. Conversion among these systems is fundamental to digital electronics.

Exercise

Convert a simple decimal number to binary.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 29: Logic Gates

Lesson Content

Basic logic gates include AND, OR, NOT, NAND, NOR, XOR, and XNOR. They implement Boolean operations used in digital circuits.

Exercise

State the truth behavior of AND, OR, and NOT gates.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 30: Boolean Algebra

Lesson Content

Boolean algebra provides mathematical rules for simplifying logic expressions. Laws such as identity, complement, distributive, and De Morgan's laws are useful in digital design.

Exercise

State two Boolean algebra laws.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 31: Combinational Logic

Lesson Content

Combinational circuits produce outputs based on current inputs. Examples include adders, multiplexers, decoders, encoders, and comparators.

Exercise

Name five combinational logic circuits.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 32: Sequential Logic

Lesson Content

Sequential circuits depend on current inputs and stored state. Flip-flops, registers, counters, and memory elements are common examples.

Exercise

Differentiate combinational and sequential logic.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 33: Flip-Flops

Lesson Content

Flip-flops are bistable circuits used to store binary state. Common types include SR, JK, D, and T flip-flops.

Exercise

Give four applications of flip-flops.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 34: Counters and Registers

Lesson Content

Counters progress through defined states, while registers store and move groups of binary data. Both are important in digital systems.

Exercise

State two applications of counters and two of registers.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 35: Sensors and Transducers

Lesson Content

A sensor detects a physical quantity, while a transducer converts energy or a physical quantity from one form to another. Examples include temperature, light, pressure, and motion sensors.

Exercise

Give five examples of sensors or transducers.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 36: Measurement Instruments

Lesson Content

Multimeters, oscilloscopes, power supplies, function generators, and specialized meters are common laboratory instruments. Correct range selection and connection are essential.

Exercise

State five electrical measurement instruments and their uses.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 37: Oscilloscope Basics

Lesson Content

An oscilloscope displays electrical signals as waveforms and helps analyze amplitude, frequency, timing, noise, and distortion.

Exercise

What can an oscilloscope reveal that a basic DC voltage reading may not?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 38: Power Supplies

Lesson Content

Power supplies provide controlled electrical energy to circuits. Important characteristics include voltage, current capacity, regulation, ripple, protection, and efficiency.

Exercise

List five characteristics of a good laboratory power supply.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 39: Electrical Safety

Lesson Content

Electrical work requires awareness of shock, burns, arc hazards, stored energy, and equipment damage. Safe practice includes isolation, suitable protective equipment, correct instruments, and following applicable procedures.

Exercise

List eight electrical safety practices.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 40: Earthing and Protection

Lesson Content

Earthing and protective devices help reduce risk from faults. Fuses, circuit breakers, residual-current protection, and suitable grounding arrangements serve different purposes.

Exercise

Explain the purpose of a fuse and a circuit breaker.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 41: Electromagnetic Induction

Lesson Content

Changing magnetic flux can induce an electromotive force in a conductor. This principle underlies transformers, generators, and many sensing systems.

Exercise

State Faraday's law conceptually.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 42: Electric Motors

Lesson Content

Motors convert electrical energy into mechanical energy. DC, induction, synchronous, and brushless motor types are used in different applications.

Exercise

Name four types of electric motors.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 43: Generators

Lesson Content

Generators convert mechanical energy into electrical energy through electromagnetic induction. They are used in power generation and backup systems.

Exercise

Explain the basic operating principle of a generator.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 44: Power Electronics

Lesson Content

Power electronics controls electrical energy using semiconductor switches and converters. Applications include motor drives, battery systems, inverters, rectifiers, and renewable-energy systems.

Exercise

Name five power-electronics applications.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 45: Control Systems

Lesson Content

Control systems regulate processes using measurements, controllers, actuators, and feedback. Open-loop systems do not use feedback, while closed-loop systems do.

Exercise

Differentiate open-loop and closed-loop control.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 46: Embedded Systems and Microcontrollers

Lesson Content

Microcontrollers combine processing, memory, and peripherals on a chip for dedicated control tasks. They are common in appliances, instruments, vehicles, and automation.

Exercise

Give five applications of microcontrollers.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 47: PCB Design and Manufacturing

Lesson Content

Printed circuit boards mechanically support and electrically connect components. Design considerations include routing, grounding, power distribution, component placement, thermal management, and manufacturability.

Exercise

List six important PCB design considerations.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 48: Electronics Troubleshooting

Lesson Content

Troubleshooting begins with symptoms, documentation, visual inspection, safe measurements, hypothesis testing, isolation, repair, and verification. Changing parts randomly is inefficient and can create additional faults.

Exercise

Write a structured electronics troubleshooting procedure.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 49: Final Revision and Engineering Practice

Lesson Content

Effective electrical and electronics engineering combines theory, measurement, design, safety, documentation, simulation, testing, and continuous improvement.

Exercise

Summarize the engineering workflow from requirements to tested circuit.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Final Assessment: 60 Questions and Answers

Attempt each question before reading the answer.

1. What is electrical engineering?

Answer: A field concerned with electrical energy, circuits, power, machines, control, and related systems.

2. What is electronics engineering?

Answer: A field focused on electronic devices and circuits used to process, control, amplify, switch, and transmit signals or energy.

3. What is voltage?

Answer: Electrical potential difference.

4. What is current?

Answer: Rate of flow of electric charge.

5. What is resistance?

Answer: Opposition to electric current.

6. What is the unit of voltage?

Answer: Volt (V).

7. What is the unit of current?

Answer: Ampere (A).

8. What is the unit of resistance?

Answer: Ohm (Ω).

9. What is the unit of power?

Answer: Watt (W).

10. State Ohm's law.

Answer: V = IR for an appropriate ohmic relationship.

11. What is electrical power?

Answer: Rate of electrical energy transfer.

12. Give a common power formula.

Answer: P = VI.

13. What is DC?

Answer: Current or voltage with a direction that does not periodically reverse in the idealized sense.

14. What is AC?

Answer: A current or voltage that periodically changes direction or magnitude.

15. What is a series circuit?

Answer: A circuit with a single current path through its series elements.

16. What is a parallel circuit?

Answer: A circuit containing branches connected across common nodes.

17. What is KCL?

Answer: Kirchhoff's Current Law: algebraic current sum at a node is zero.

18. What is KVL?

Answer: Kirchhoff's Voltage Law: algebraic voltage sum around a closed loop is zero.

19. What is a resistor?

Answer: A component used to provide electrical resistance.

20. Why is resistor power rating important?

Answer: It indicates the amount of power the resistor can safely dissipate under specified conditions.

21. What is a capacitor?

Answer: A component that stores energy in an electric field.

22. What is an inductor?

Answer: A component that stores energy in a magnetic field.

23. What is a transformer?

Answer: A device that transfers AC energy between circuits through electromagnetic induction.

24. What is a diode?

Answer: A semiconductor device designed to conduct preferentially in one direction under normal operation.

25. Why does an LED require current control?

Answer: Excess current can damage it, so a suitable circuit limits or regulates current.

26. What is a rectifier?

Answer: A circuit that converts AC into pulsating DC.

27. What is a transistor?

Answer: A semiconductor device used for switching, amplification, or signal control.

28. Name three transistor applications.

Answer: Switching, amplification, and regulation/control.

29. What is a MOSFET?

Answer: A field-effect transistor controlled primarily by an electric field at its gate.

30. What is an operational amplifier?

Answer: A high-gain differential amplifier commonly used with feedback.

31. What is negative feedback?

Answer: Returning part of an output signal in a way that reduces the effective input error or gain.

32. What is a digital signal?

Answer: A signal represented using discrete states or levels.

33. What is binary?

Answer: A base-2 number system using 0 and 1.

34. Name seven basic logic gates.

Answer: AND, OR, NOT, NAND, NOR, XOR, and XNOR.

35. What is Boolean algebra?

Answer: A mathematical system for representing and simplifying logical relationships.

36. What is a combinational circuit?

Answer: A digital circuit whose output depends on current inputs.

37. What is a sequential circuit?

Answer: A circuit whose output depends on inputs and stored state.

38. What is a flip-flop?

Answer: A bistable digital circuit capable of storing one bit of state.

39. What is a sensor?

Answer: A device or element that detects a physical quantity.

40. What is a transducer?

Answer: A device that converts energy or a physical quantity between forms.

41. What is a multimeter?

Answer: An instrument used to measure quantities such as voltage, current, and resistance.

42. What is an oscilloscope?

Answer: An instrument that displays electrical waveforms versus time.

43. What does a laboratory power supply do?

Answer: Provides controlled electrical power to a circuit under specified limits.

44. Why is electrical safety important?

Answer: Electrical energy can cause serious injury, equipment damage, fire, or other hazards.

45. What is a fuse?

Answer: A protective device designed to interrupt excessive current by opening the circuit.

46. What is a circuit breaker?

Answer: A resettable or serviceable protective switching device that interrupts abnormal current conditions.

47. What is electromagnetic induction?

Answer: Generation of electromotive force due to changing magnetic flux.

48. What does an electric motor do?

Answer: Converts electrical energy into mechanical energy.

49. What does a generator do?

Answer: Converts mechanical energy into electrical energy.

50. What is power electronics?

Answer: The control and conversion of electrical energy using power semiconductor devices.

51. What is an open-loop control system?

Answer: A control system without feedback from the output.

52. What is a closed-loop control system?

Answer: A control system that uses feedback to influence control action.

53. What is a microcontroller?

Answer: An integrated circuit containing a processor, memory, and peripherals for dedicated control applications.

54. What is a PCB?

Answer: A printed circuit board that supports and electrically connects electronic components.

55. Why is PCB grounding important?

Answer: Good grounding helps provide predictable return paths and can reduce noise and interference.

56. What is troubleshooting?

Answer: A systematic process for locating and correcting faults.

57. Why measure before replacing components?

Answer: Measurements provide evidence and help identify the actual fault.

58. What is a circuit schematic?

Answer: A symbolic diagram showing electrical connections and components.

59. Why is documentation important in engineering?

Answer: It supports design review, testing, maintenance, repeatability, and future troubleshooting.

Practical Exercises

Exercise 1 — Basic Circuit Analysis

Construct or simulate a simple resistor circuit only under appropriate supervision and safe low-voltage conditions. Measure voltage and current and compare your observations with Ohm's law.

Exercise 2 — Component Identification

Identify resistors, capacitors, diodes, LEDs, transistors, switches, connectors, and other components from a laboratory component set. Record their symbols and typical functions.

Exercise 3 — Digital Logic

Create truth tables for AND, OR, NOT, NAND, NOR, XOR, and XNOR gates. Then simplify one Boolean expression using appropriate laws.

Exercise 4 — Measurement Practice

Using a suitable low-voltage training setup and under supervision, practice measuring DC voltage, resistance, and continuity with a multimeter. Record the measurement range and result.

Exercise 5 — Troubleshooting

Create a fault-finding procedure for a circuit that has no output. Include visual inspection, power checks, continuity checks, signal checks, isolation, repair, and final verification.

Electrical and Electronics Engineering Glossary

Voltage: Electrical potential difference measured in volts.

Current: Rate of electric charge flow measured in amperes.

Resistance: Opposition to current measured in ohms.

Power: Rate of energy transfer measured in watts.

Energy: Capacity to perform work, often measured in joules or watt-hours.

Resistor: Component providing electrical resistance.

Capacitor: Component storing energy in an electric field.

Inductor: Component storing energy in a magnetic field.

Diode: Semiconductor device with directional conduction behavior.

LED: Light-emitting diode.

Transistor: Semiconductor device used for switching, amplification, and control.

MOSFET: Metal-oxide-semiconductor field-effect transistor.

Rectifier: Circuit converting AC to pulsating DC.

Transformer: Electromagnetic device transferring AC energy between circuits.

Amplifier: Circuit that increases signal power or amplitude under defined conditions.

Oscilloscope: Instrument for observing electrical waveforms.

Multimeter: Instrument for measuring common electrical quantities.

Sensor: Device that detects a physical quantity.

Transducer: Device converting one form of energy or physical quantity to another.

PCB: Printed circuit board.

HACCP: Not applicable to electrical engineering; this term belongs to food safety.

Microcontroller: Integrated processor, memory, and peripherals for dedicated control.

Feedback: Using information about system output to influence control action.

Fuse: Overcurrent protective device that opens a circuit when its element operates.

Circuit breaker: Protective switching device that interrupts abnormal current conditions.

End of Electrical and Electronics Engineering

Strong engineering practice combines theory, safe measurement, sound design, careful documentation, systematic troubleshooting, and verification. Practical work should always follow appropriate safety procedures and supervision.